

surveydown

An Open-Source, Markdown-Based Platform
for Reproducible and Programmable Surveys

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Quant UX Con 2025



About Me



- My name is **Pingfan Hu**, a PhD Candidate at George Washington University, supervised by Dr John Helveston.
- Graduating in **Spring 2025**.
- Research focuses:
 1. Sustainable transportation
 2. Choice modeling & data analysis
 3. Research software development
- For more information, visit pingfanhu.com

I try to figure out what people want...





Prof. John
Helveston



jhelvy.shinyapps.io



Welcome to the Ultimate Superhero Survey!



Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

Do you have a favorite super hero? *

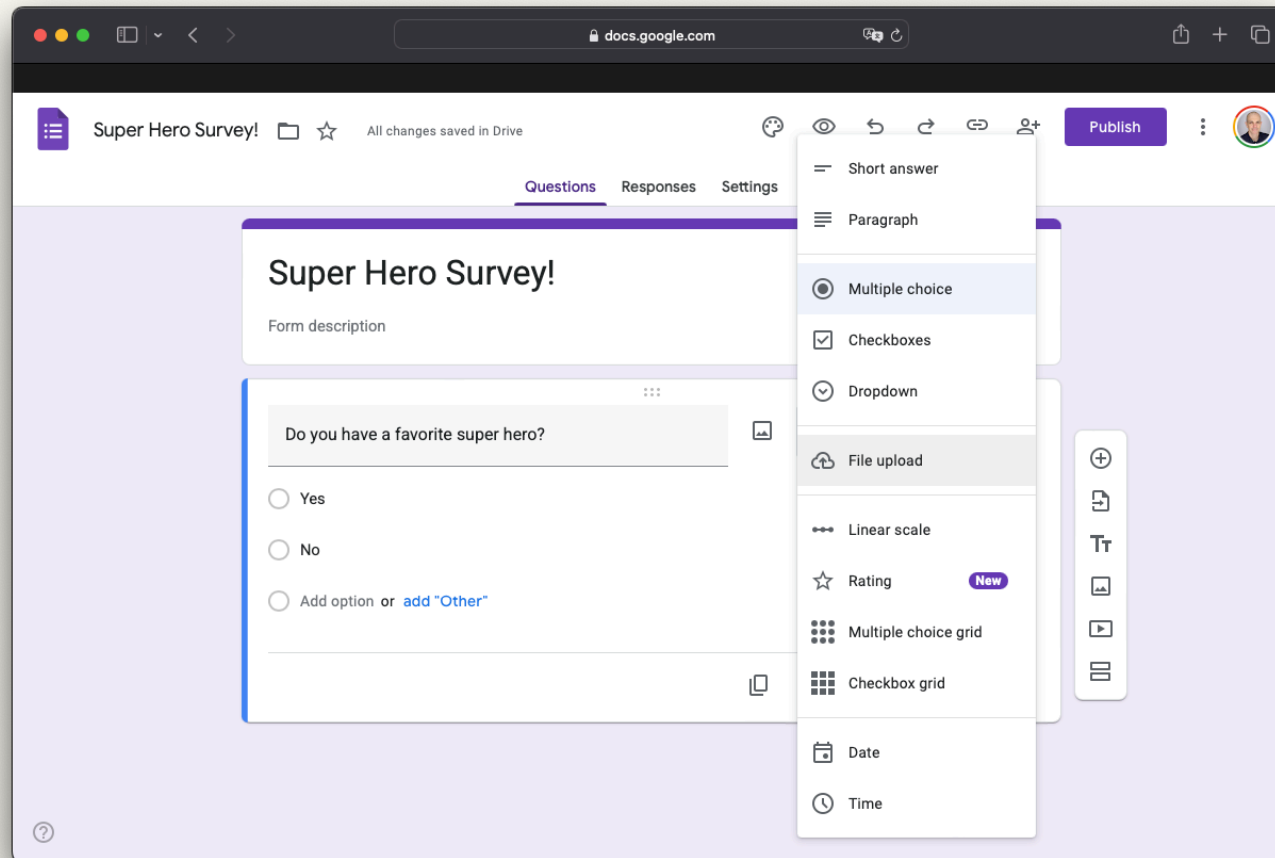
☐ Yes

☒ No

Next

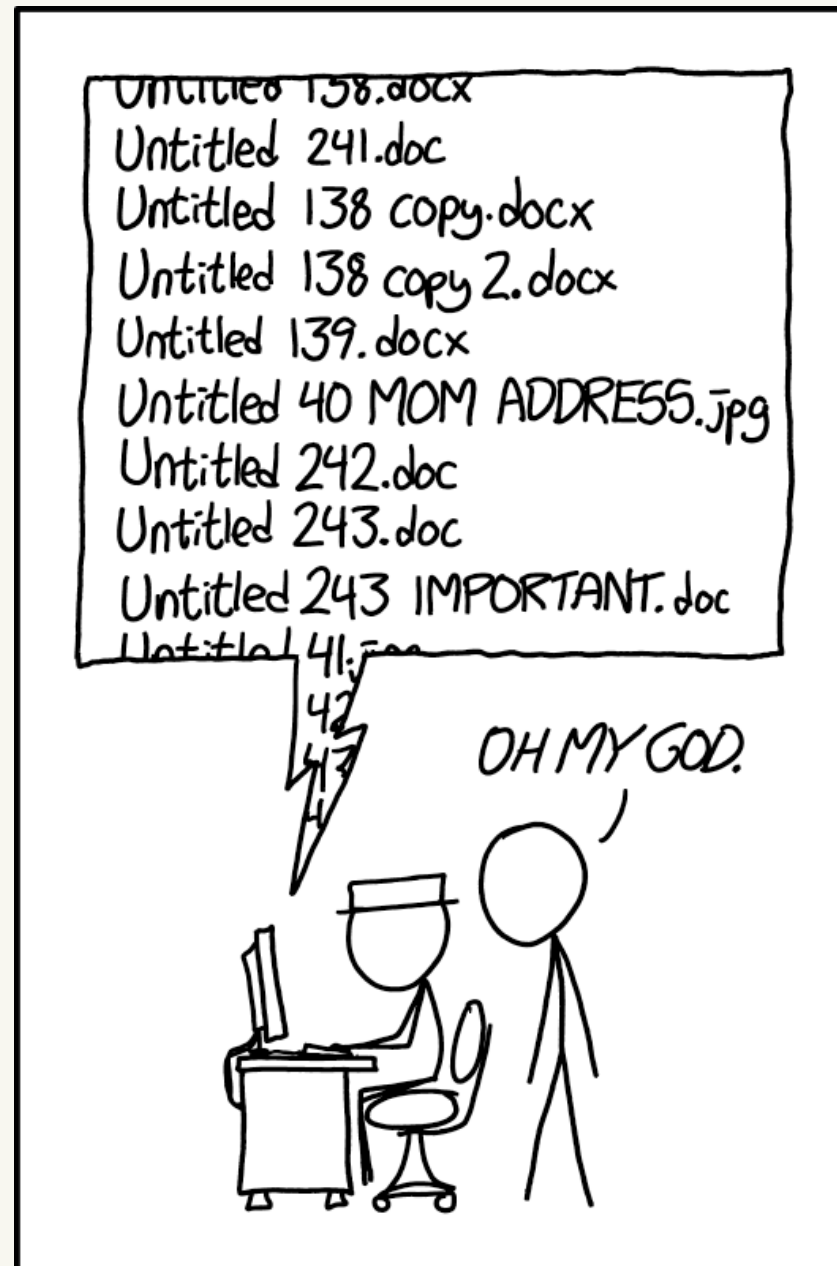
WYSIWYG Interface

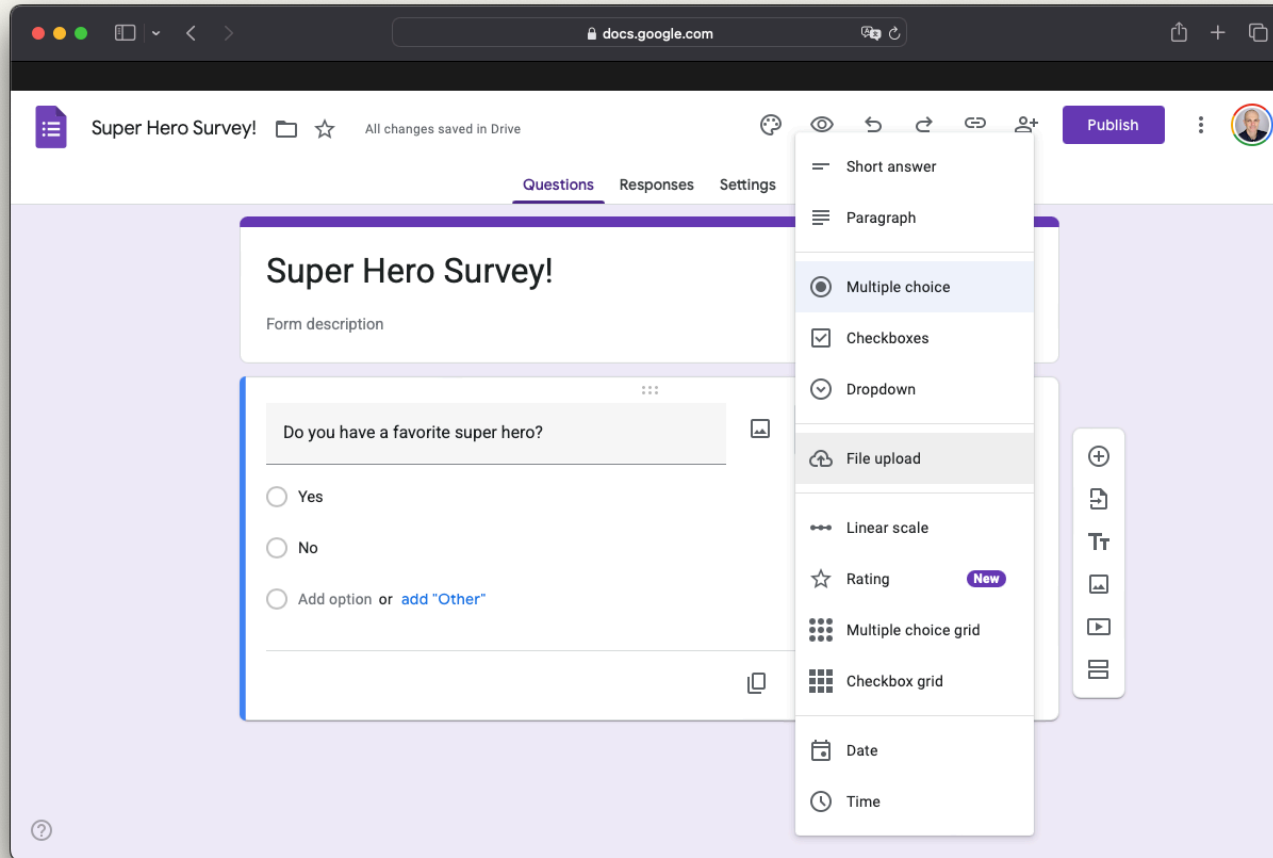
The image shows a web browser window displaying the Google Forms WYSIWYG editor. The browser's address bar shows 'docs.google.com'. The form is titled 'Super Hero Survey!' and has a subtitle 'Form description'. The 'Questions' tab is selected in the top navigation bar. A question is being edited: 'Do you have a favorite super hero?'. Below the question, there are three radio button options: 'Yes', 'No', and 'Add option or add "Other"'. A context menu is open on the right side of the question, listing various question types: 'Short answer', 'Paragraph', 'Multiple choice' (which is highlighted), 'Checkboxes', 'Dropdown', 'File upload', 'Linear scale', 'Rating' (marked with a 'New' badge), 'Multiple choice grid', 'Checkbox grid', 'Date', and 'Time'. On the far right, there is a vertical toolbar with icons for adding new questions, inserting a section, changing the theme, adding an image, adding a video, and adding a link. The top right of the interface includes a 'Publish' button and a user profile icon.



Limitations

- ✗ Reproducibility
- ✗ Version control





Limitations

- ✗ Reproducibility
- ✗ Version control
- ✗ Limited features
- ✗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



In this talk, we will cover:

What is surveydown?

How does it work?

What can I do with it?

What's next?

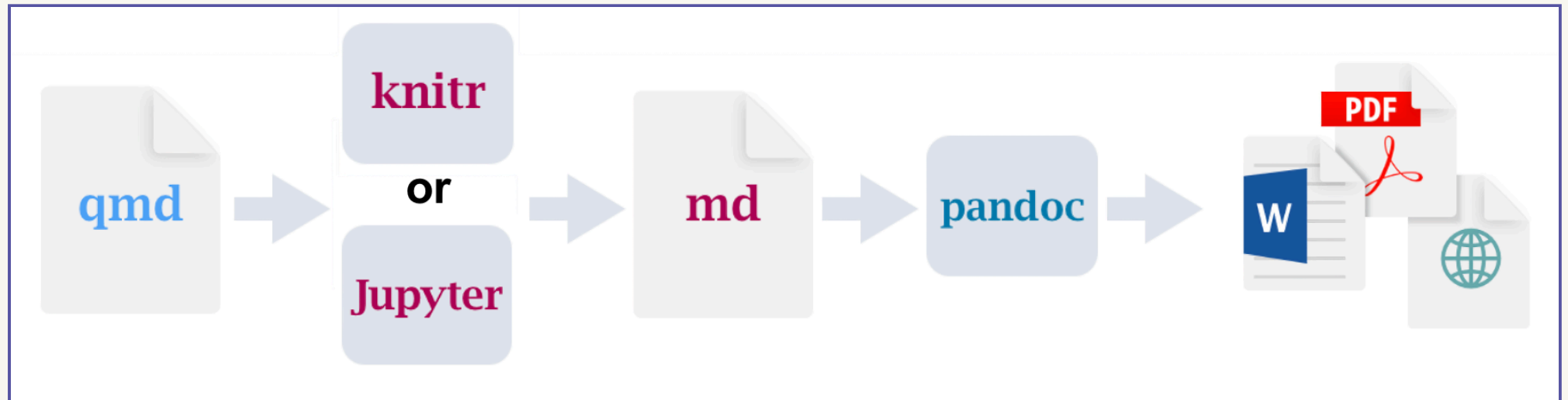
What is **surveydown**?



R package that renders Quarto files into surveys



Quarto is a publishing system

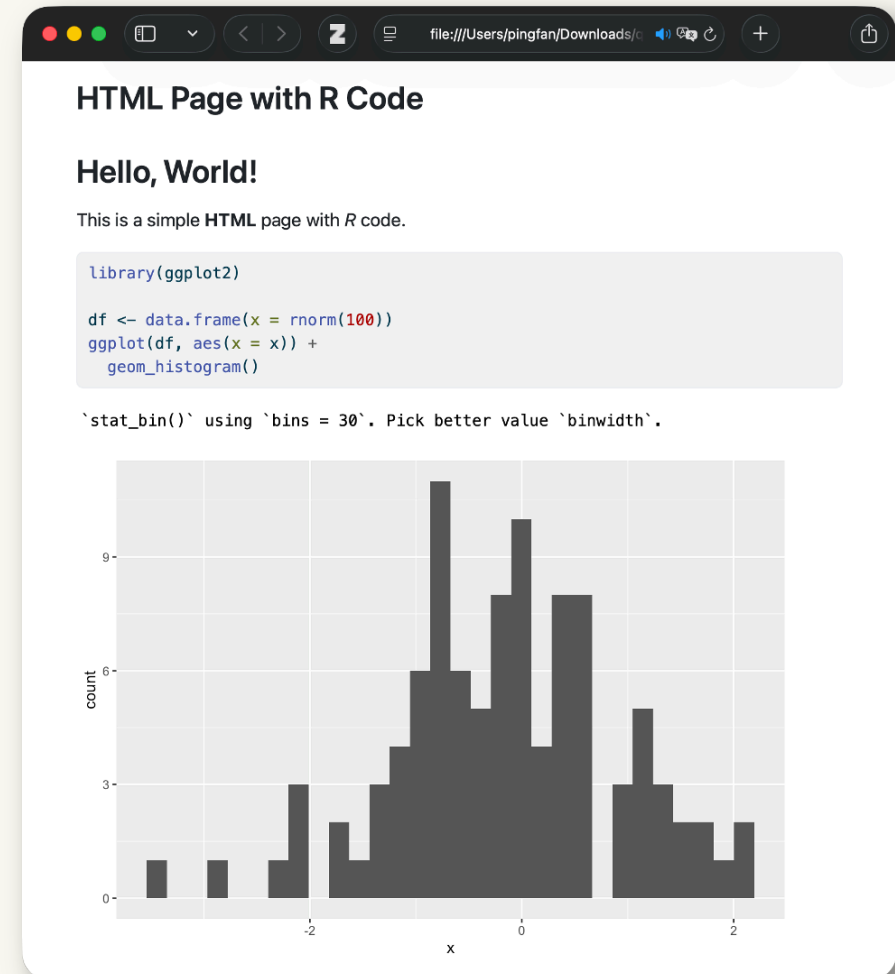


Original qmd file

Markdown + R code chunks

```
1 ---
2 format: html
3 title: "HTML Page with R Code"
4 ---
5
6 # Hello, World!
7
8 This is a simple **HTML** page with *R* code.
9
10 ```{r}
11 library(ggplot2)
12
13 df <- data.frame(x = rnorm(100))
14 ggplot(df, aes(x = x)) +
15   geom_histogram()
16 ```
```

Rendered HTML

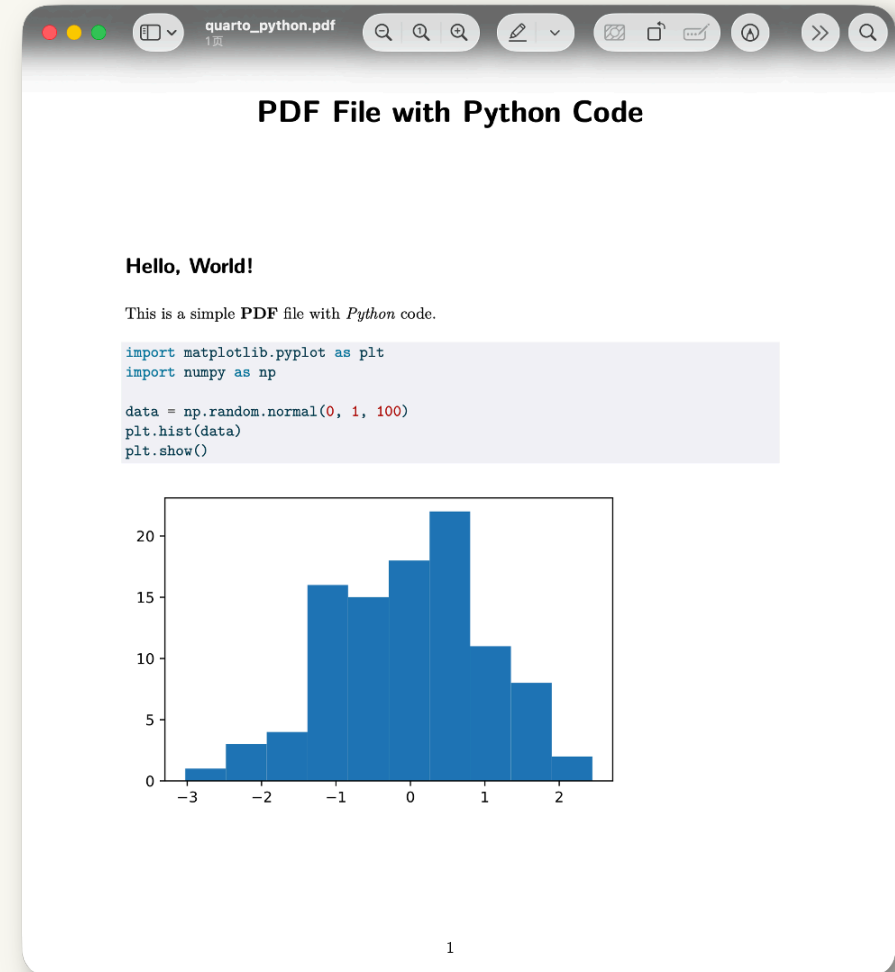


Original qmd file

Markdown + Python code chunks

```
1 ---
2 format: pdf
3 title: "PDF File with Python Code"
4 ---
5
6 # Hello, World!
7
8 This is a simple **PDF** file with *Python* code.
9
10 ```{python}
11 import matplotlib.pyplot as plt
12 import numpy as np
13
14 data = np.random.normal(0, 1, 100)
15 plt.hist(data)
16 plt.show()
17 ```
```

Rendered PDF

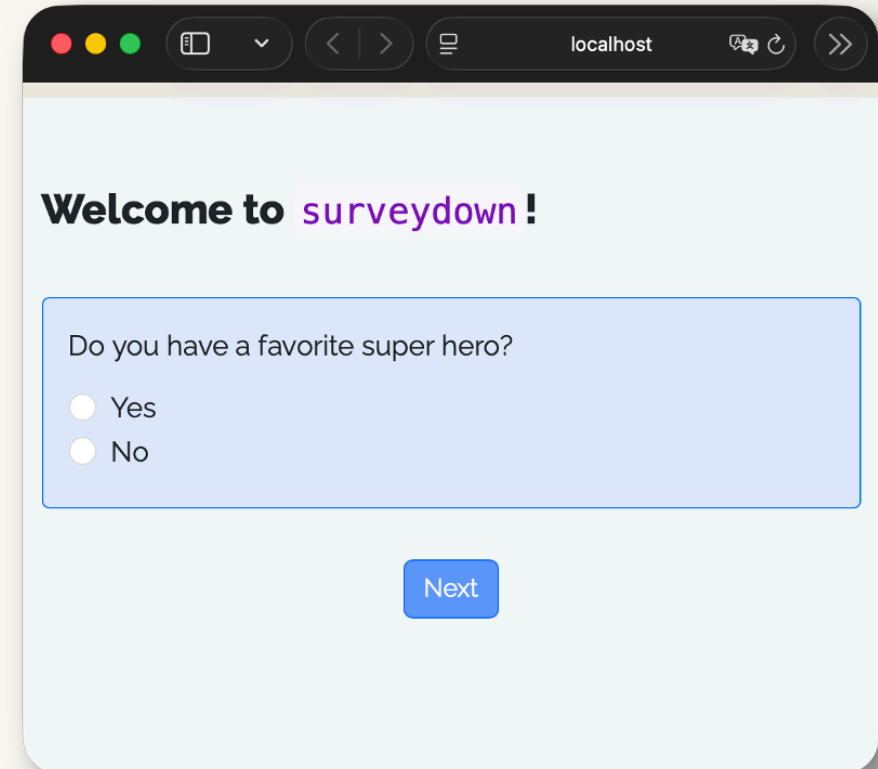




survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Rendered Survey



localhost

Welcome to surveydown !

Do you have a favorite super hero?

☐ Yes

☐ No

Next

survey.qmd

YAML header for a “clean” output

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id   = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Load the **surveydown** Package

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Use triple-dash marker (---) to define survey pages

Quarto fences (:::) also OK:

```
::: {#welcome .sd-page}
```

Page content

```
:::
```

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

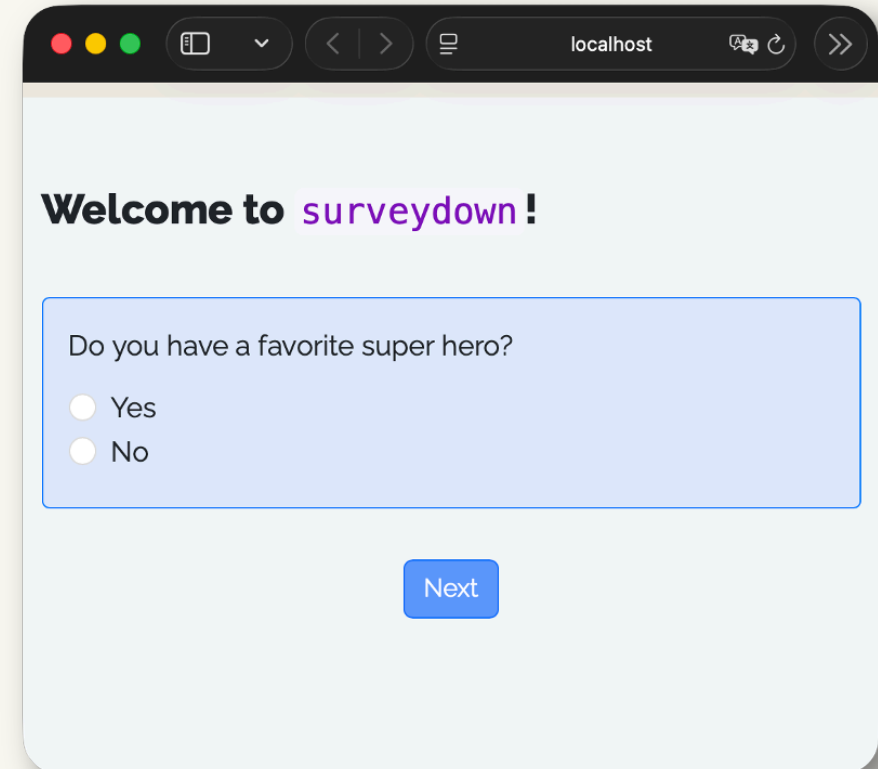
Page content

- Markdown for texts, images, etc.
- `sd_question()` for survey questions
- `sd_next()` for page navigation

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Rendered Survey



localhost

Welcome to **surveydown** !

Do you have a favorite super hero?

☐ Yes

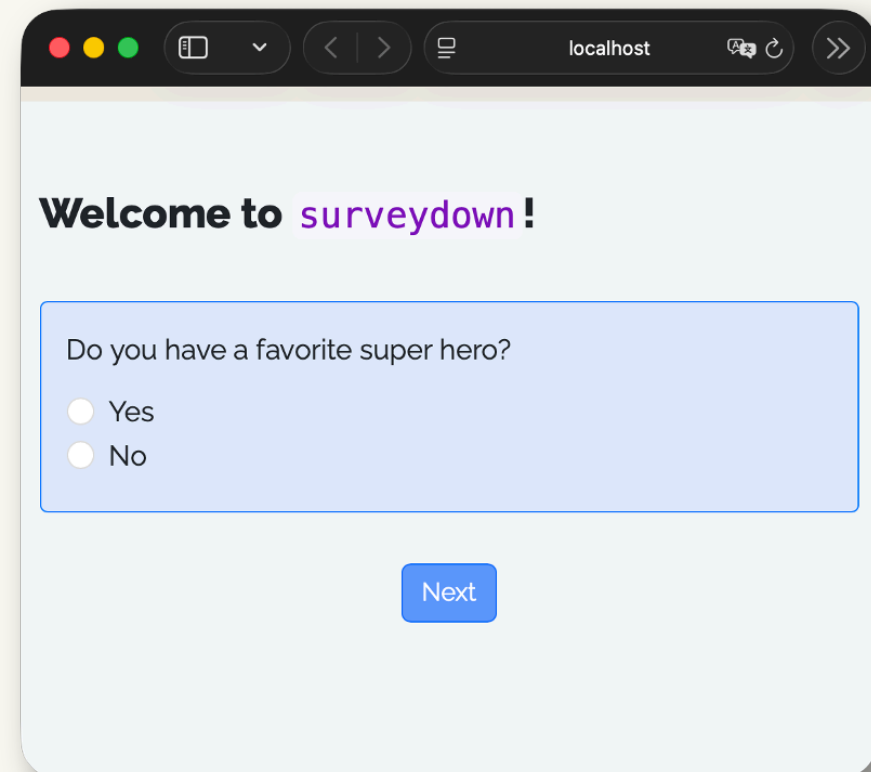
☐ No

Next

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25
26 sd_next()
27 ```
28
29 --- some_other_page
30
31 Other content...
```

Rendered Survey



Welcome to surveydown!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

has_fav_hero
yes
no
no
yes

Wait a minute...

Quarto renders to **static** html pages, right?



Shiny to the rescue!





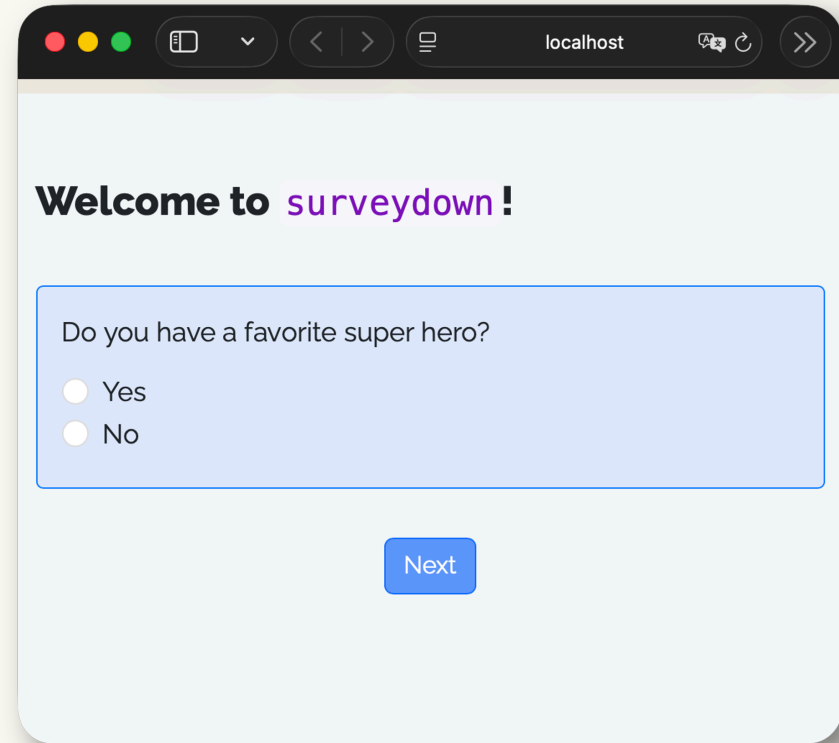
Shiny App



UI



=

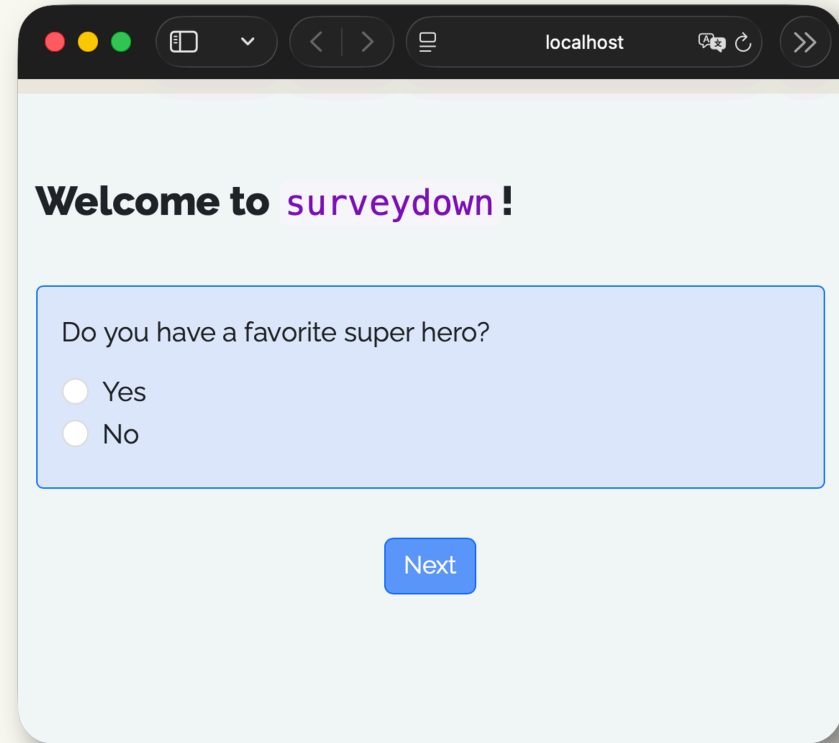


Shiny App

UI



Server



A complete surveydown survey

`survey.qmd`

A **Quarto doc** defining the survey content (pages, texts, images, questions, etc).

`app.R`

An **R script** defining the survey Shiny app.

Don't panic!



app.R

```
1 library(surveydown)
2
3 ui ← sd_ui()
4
5 server ← function(input, output, session) {
6
7   sd_server()
8
9 }
10
11 shiny::shinyApp(ui = ui, server = server)
```

app.R

```
1 library(surveydown)
2
3 ui ← sd_ui()
4
5 server ← function(input, output, session) {
6
7   sd_server()
8
9 }
10
11 shiny::shinyApp(ui = ui, server = server)
```

Render `survey.qmd` as UI

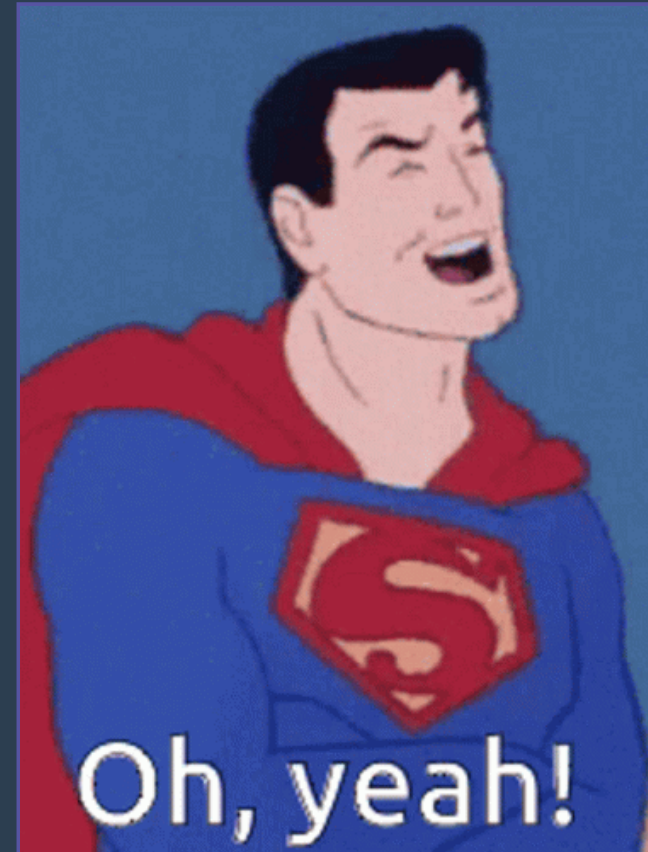
Run the `surveydown` server

Wait a minute...

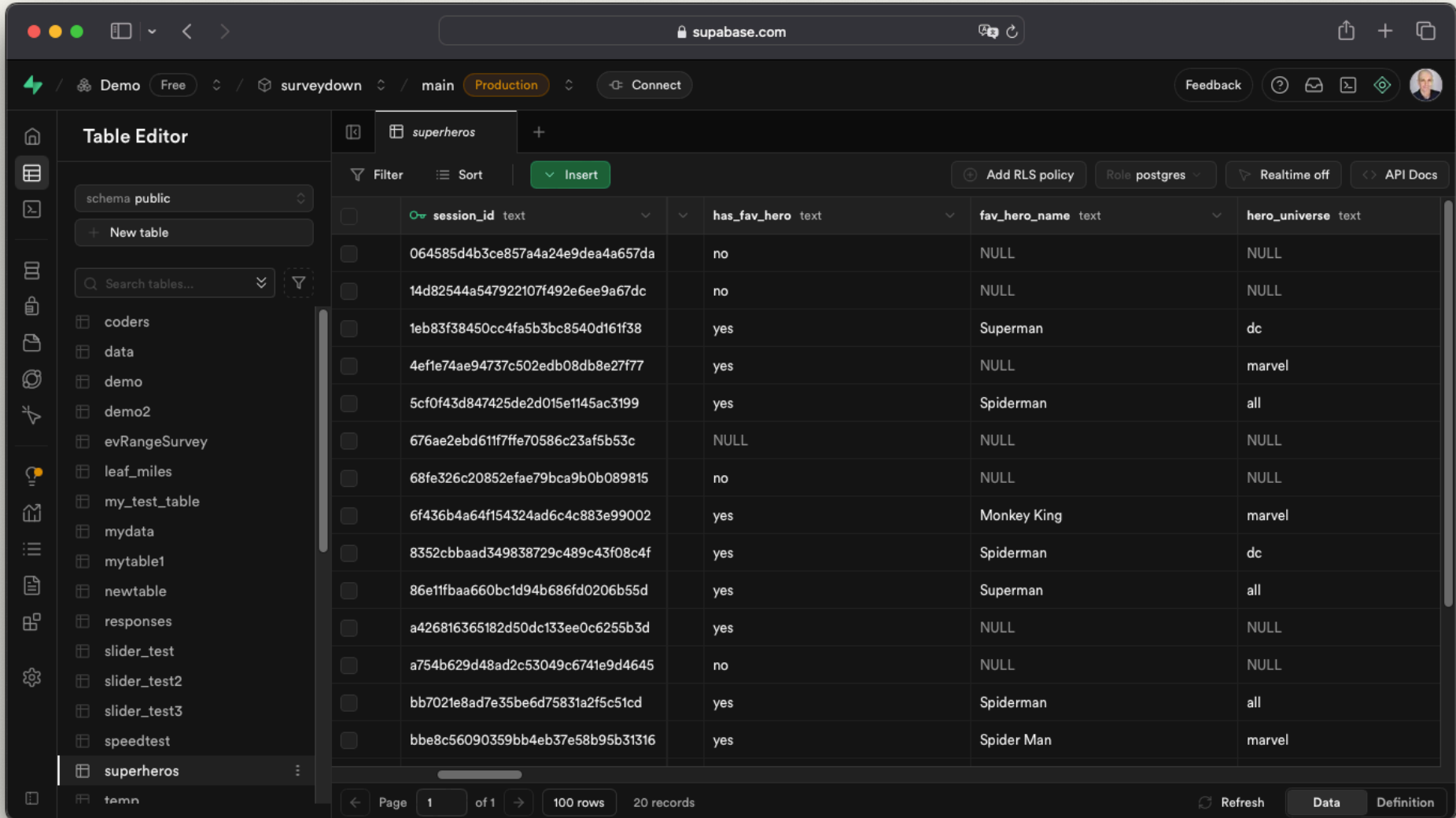
How do you store the response **data**?



PostgreSQL to the rescue!



supabase.com



app.R

```
1 library(surveydown)
2
3 # sd_db_config()
4 db <- sd_db_connect()
5
6 ui <- sd_ui()
7
8 server <- function(input, output, session) {
9
10   sd_server(db)
11
12 }
13
14 shiny::shinyApp(ui = ui, server = server)
```

Store credentials
Connect to the database

Pass connection to
sd_server()

```
demo - main - RStudio

survey.qmd x app.R x analysis.R x
Source on Save Run Source

1 library(surveydown)
2 library(dplyr)
3
4 db <- sd_db_connect()
5
6 data <- sd_get_data(db = db)
7
8 glimpse(data)
9

Console Terminal Background Jobs
R 4.4.1 ~ /gh/slides/2025-posit-conf-surveydown/demo/

> db <- sd_db_connect()
✓ Successfully connected to the database.
>
> data <- sd_get_data(db = db)
>
> glimpse(data)
Rows: 20
Columns: 50
$ session_id <chr> "5cf0f43d847425..."
$ time_start <chr> "2025-08-15 15:..."
$ time_end <chr> "2025-08-15 15:..."
$ completion_code <chr> "37081685", "60..."
$ browser <chr> "Safari v26, iO..."
$ ip_address <chr> "140.248.0.0", ...
$ has_fav_hero <chr> "yes", "yes", "..."
$ fav_hero_name <chr> "Spiderman", "S..."
$ hero_universe <chr> "all", "dc", "a..."
$ hero_qualities <chr> "strength, flyi..."
$ dream_power <chr> "telepathy", "t..."
$ responsibility_scale <chr> "10", "8.1", "1..."
$ villain_encounter <chr> "help_backup", ...
$ superhero_rating <chr> NA, NA, NA, NA, ...
$ superhero_rating_spiderman <chr> "love", "love", ...
$ superhero_rating_batman <chr> "love", NA, "li..."
$ superhero_rating_wonder_woman <chr> "love", NA, "li..."

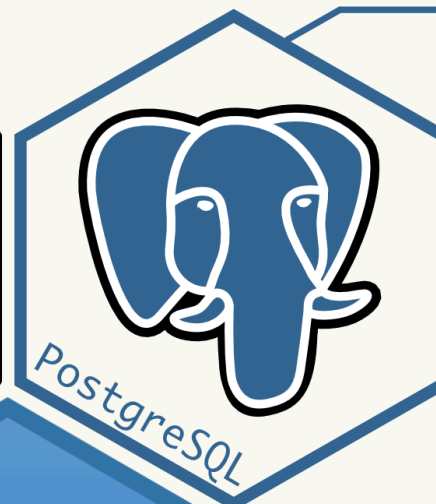
9:1 (Top Level) Copilot: Completion response received. R Script
Environment History Connections Git Tutorial Files Plots Packages Help Viewer Presentation
```

What is **surveydown**?



Control Logic

Data Storage



Survey Design

Web Framework

What can **surveydown** do?

surveydown is feature-packed!

surveydown.org

AboutDocumentationTemplatesBlog

FAQ

Documentation

Features & Roadmap

Survey Basics

Getting Started

Basic Components

Question Development

Defining Questions

Question Types

Question Formatting

Custom Questions

Survey Design Concepts

Survey Settings

Page Navigation

Conditional Logic

System Translations

External Resources

Interactivity

Reactivity

Randomization

External Redirect

Data Management

Storing Data

Fetching Data

Features

Feature	Docs	Version	Discussion
Wide variety of question types	Question Types	v0.0.1	105 , 109
Conditionally show questions	Conditional Display	v0.0.1	
Markdown formatting for options and buttons	Markdown Formatting	v0.0.1	
Require specific questions or all questions be answered	Required Questions	v0.0.2	
Support for bootstrap themes	Themes	v0.0.4	
Customizable scss theme file	Themes	v0.0.4	
Ability to Ignore the database connection	Supabase Ignore	v0.0.8	
Time stamps recorded for each question and page interaction		v0.0.9	
Progress bar that updates on each question interaction	Progress Bar	v0.0.9	
Customizable progress bar color and position on page	Progress Bar	v0.0.9	
Ability to use latest survey results in the survey itself	Fetching Data	v0.1.1	
Pass parameters through the url	Reactive	v0.2.2	92

On this page

surveydown package

Features

To Do

sdstudio package

Features

To Do

Site made with   and  quarto

Feature Highlights

Question types

Conditional logic

12 Built-in Question Types

text

textarea

numeric

mc

mc_multiple

mc_buttons

mc_multiple_buttons

select

slider

slider_numeric

date

daterange

Question Type: text

```
1 sd_question(  
2   type  = "text",  
3   id    = "fav_hero_name",  
4   label = "Who is your favorite super hero?"  
5 )
```



Who is your favorite super hero?

Question Type: mc_buttons

```
1 sd_question(  
2   type = "mc_buttons",  
3   id   = "dream_power",  
4   label = "If you could have ONE superpower?",  
5   option = c(  
6     "🕸 Web-slinging" = "webslinging",  
7     "🛡 Super Strength" = "strength",  
8     "✈ Flight" = "flight",  
9     "🧠 Telepathy" = "telepathy",  
10    "⚡ Super Speed" = "speed"  
11  ),  
12  direction = "vertical"  
13 )
```



If you could have ONE superpower?

 Web-slinging Swing through cities
 Super Strength Lift anything
 Flight Soar through the skies
 Telepathy Read minds
 Super Speed Faster than lightning

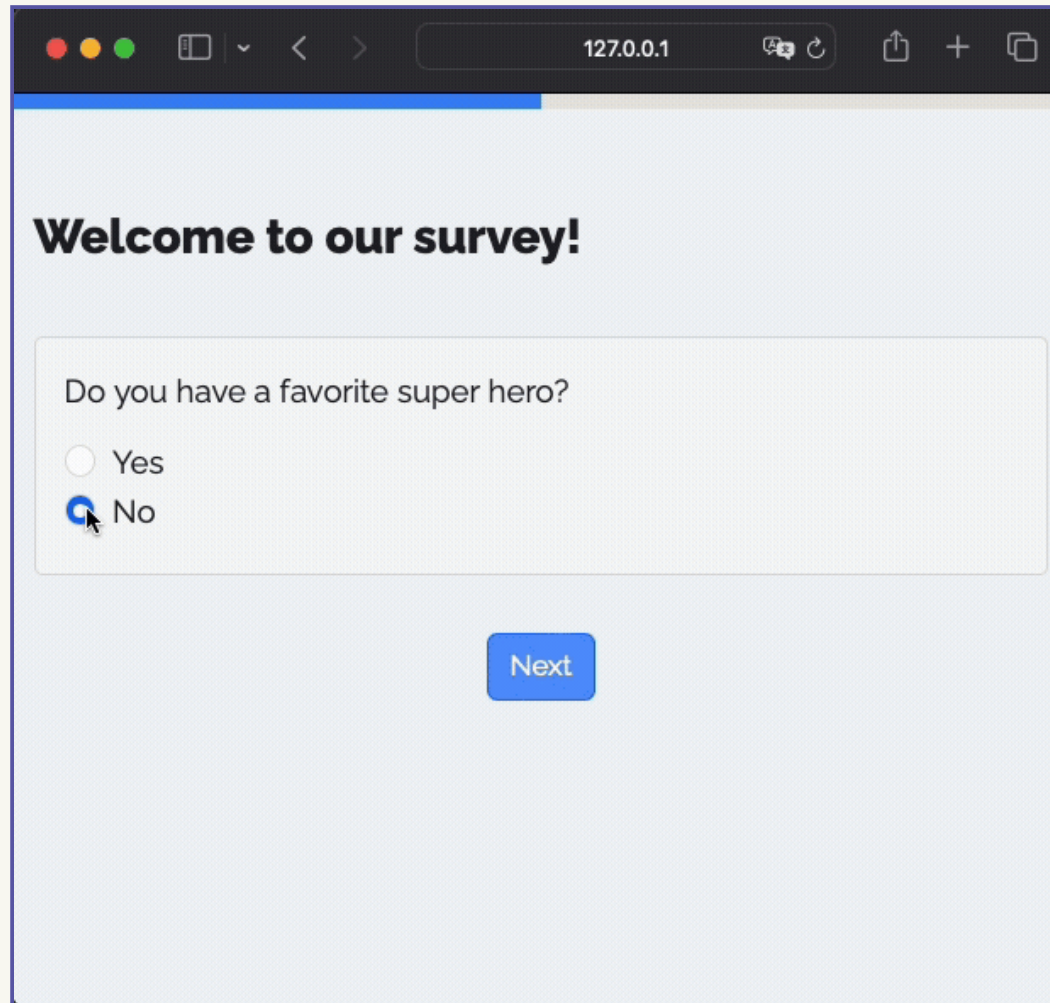
Conditional logic

Conditional showing

Conditional skipping

Conditional stopping

Conditional showing - `sd_show_if()`



A screenshot of a web browser window displaying a survey. The browser's address bar shows the URL `127.0.0.1`. The survey content is as follows:

Welcome to our survey!

Do you have a favorite super hero?

☐ Yes

☒ No

Next

Conditional showing - `sd_show_if()`

survey.qmd

```
1 # Conditional Question
2 sd_question(
3   type = "mc",
4   id    = "has_fav_hero",
5   label = "Do you have a favorite hero?",
6   option = c("Yes" = "yes", "No" = "no")
7 )
8
9 # Target Question
10 sd_question(
11   type = "text",
12   id    = "fav_hero",
13   label = "Who is your favorite super hero?"
14 )
```

app.R

```
1 # Inside server
2 sd_show_if(
3   input$has_fav_hero == "yes" ~ "fav_hero"
4 )
```

Condition ~ Target

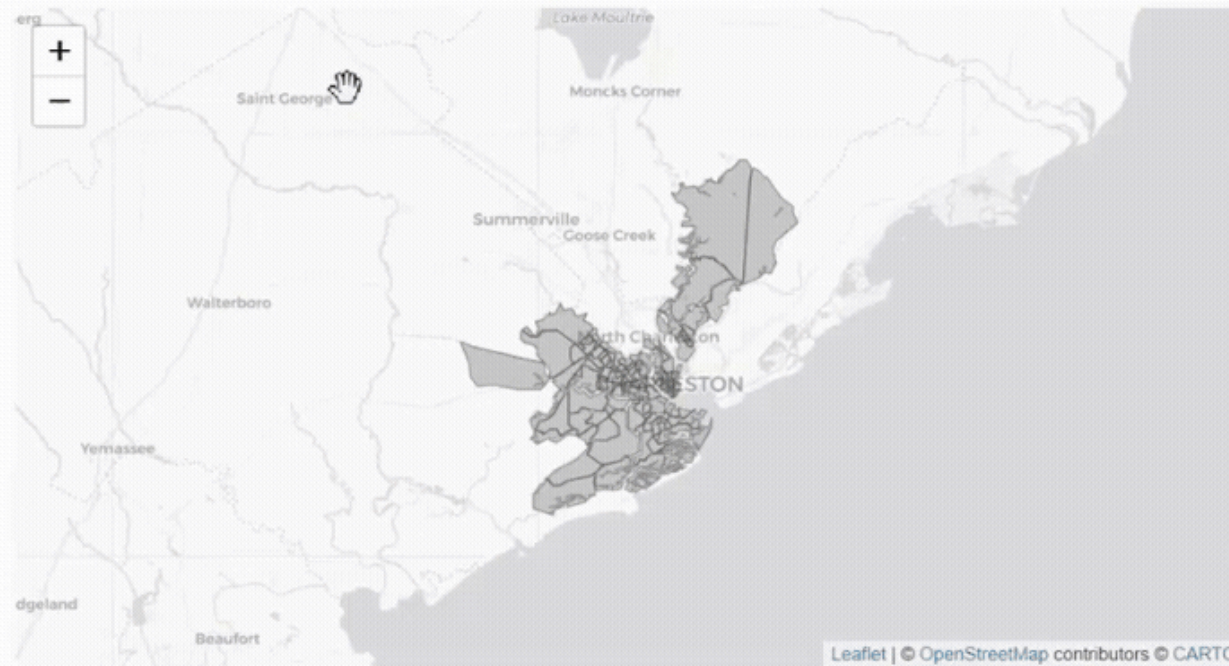
“If {condition}, then show {target}”

If it works in **Shiny**,
it works in **surveydown**.

Embedding an interactive map with leaflet:

Question 1: Social Vulnerability is how likely people are to face greater harm or impact on their lives after disruptive hazard events like flooding based on social or economic conditions like income or transportation access.

Please review the map of the City of Charleston, South Carolina, and select the areas you believe are more socially vulnerable to the impacts of hazardous events.

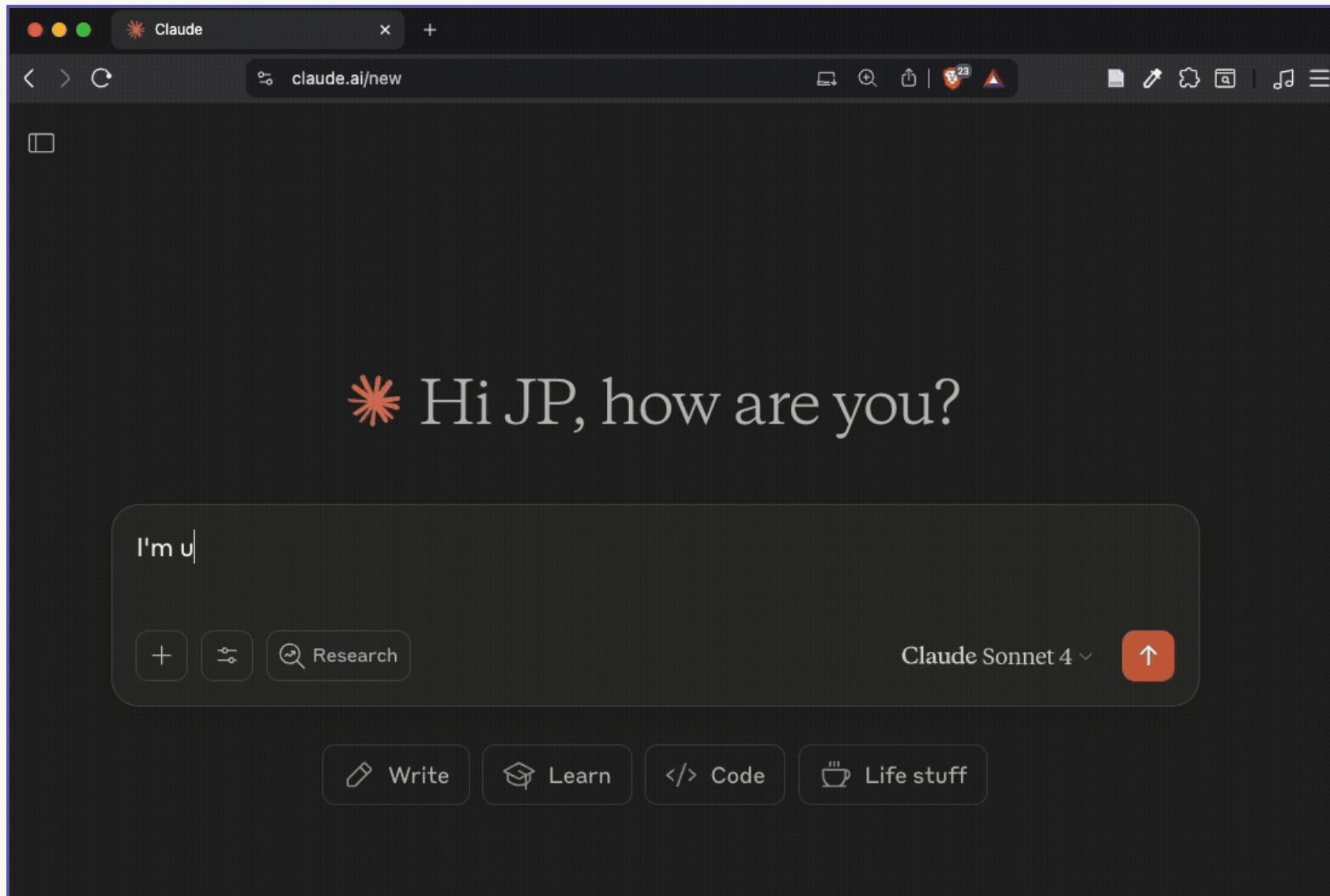


Next Question



surveydown + LLMs
is pretty cool

Generate surveys with LLMs!



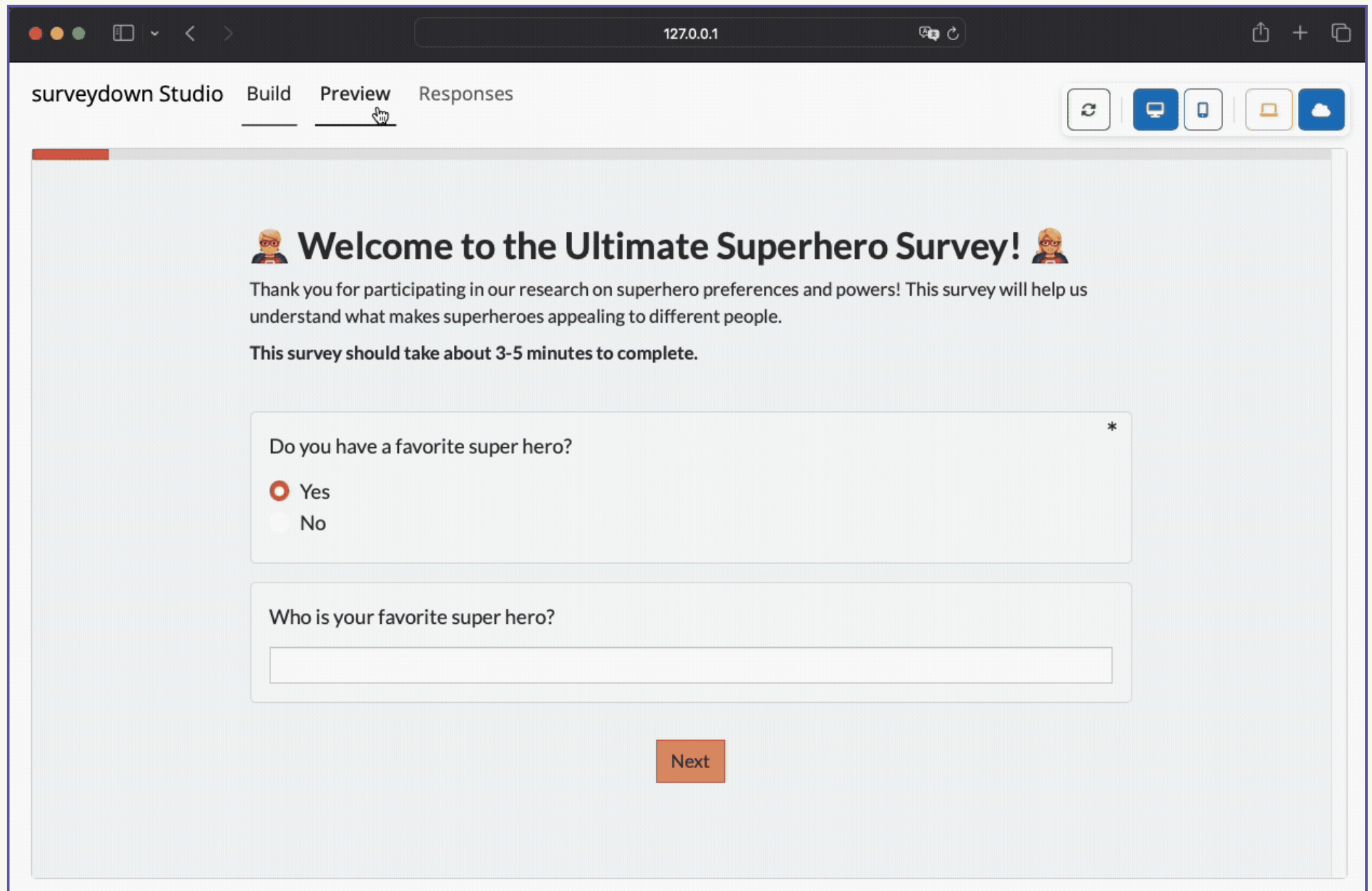
What's next?

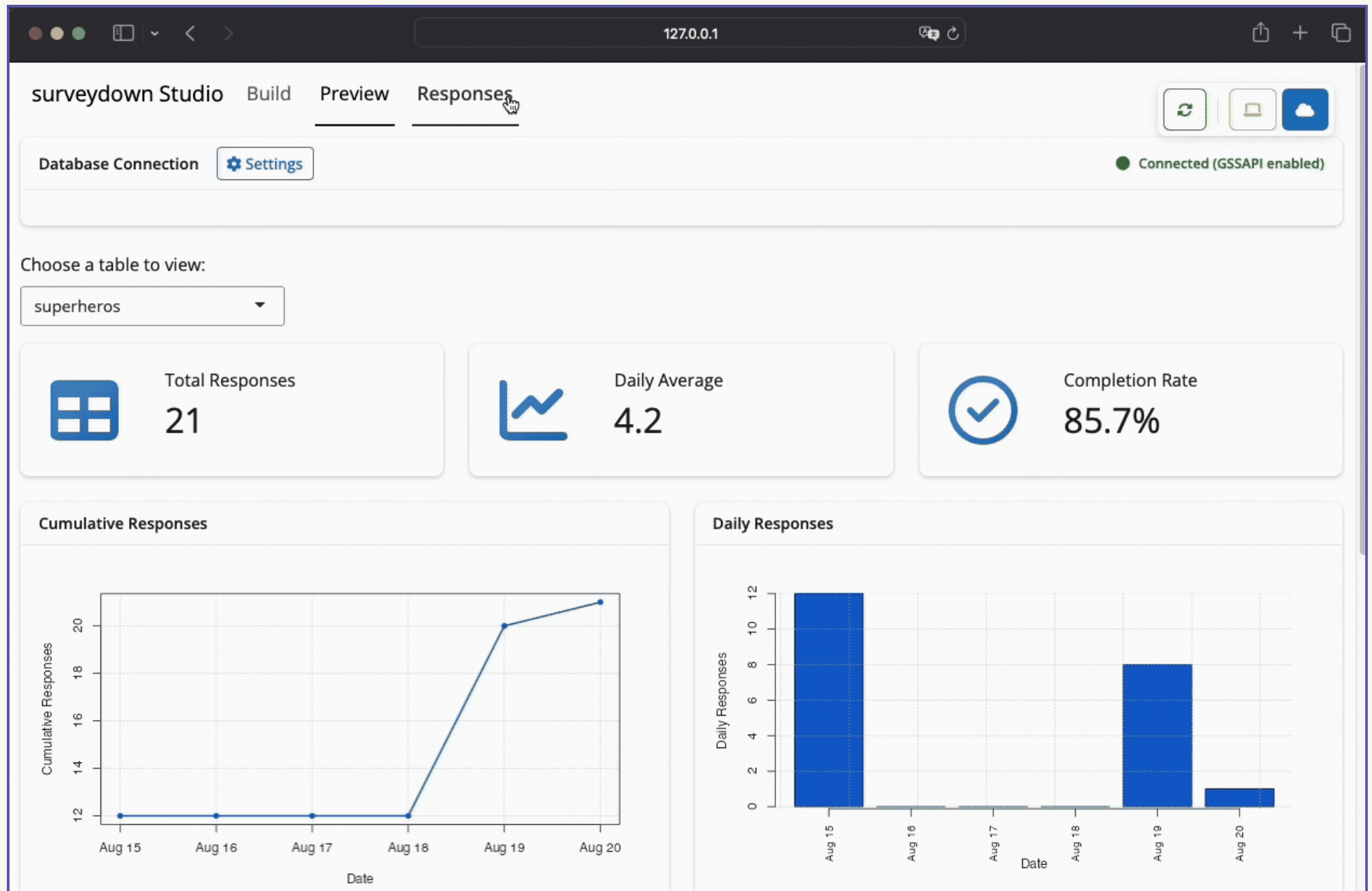
Try out the companion `sdstudio` package:

```
sdstudio::launch()
```

The screenshot displays the surveydown Studio application window. The top navigation bar includes tabs for "surveydown Studio", "Build", "Preview", and "Responses". Below this, there are two main panels:

- Structure Panel (Left):** This panel shows a hierarchical view of the survey pages. It lists three pages: "Page: welcome", "Page: hero_details", and "Page: powers". Each page has associated questions or sections listed below it. For example, under "Page: welcome", there is a title section "# Welcome to the..." and two multiple-choice questions: "has_fav_hero | mc" and "fav_hero_name | text".
- Code Panel (Right):** This panel shows the R code used to generate the survey. The code is organized into sections corresponding to the survey elements. It starts with a header section, followed by a library load for "surveydown". Then, it defines the first page with its ID and title. The code for the "has_fav_hero" question is shown, including its type ("mc"), ID, label, and options.





**With great power comes great
responsibility.**

Ways to help:

github.com/surveydown-dev/surveydown/

- Try it out!
- Give it a ★
- Post an issue
- Join the GitHub discussion
- Contribute a template



Meet the team!

surveydown.org



John Helveston, Ph.D.



Pingfan Hu



Bogdan Bunea



One more thing...

Workshop by Dr John Helveston Tomorrow

jhelvy.github.io/2025-qux-surveydown-workshop/

Thursday Nov 6, 2:00 PM UTC – 5, Eastern Time (U.S.)